Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element I Initial Template

How to use this template

You must test *all* high priority requirements to score a 5. After testing, you will connect the data you collected to the "success" or "failures" of your design. You will consider next steps for your design based on the data you just analyzed. Each test that was listed in your table for Element H (testing plan) was on its own row. Generate a report for each test, the results, and its analysis that includes the following:

a) number and name of the test (as referenced in Element H)
b) indication of whether the test is high, medium, or low priority based on your requirements (as reference in Element C)
c) for physical tests, include an explanation of your process and the full set of raw data such that the following best practices for scientific testing and mathematical modeling will be evident:
   - creating experimental controls
   - defining independent and dependent variables
   - only changing one variable at a time and changing it in increments to find a trend
   - collecting at least FIVE trials for each variable increment
   - plotting data appropriately (e.g. scatter plot with trendline for continuous variables, bar charts/histogram for non-continuous variables, acknowledging and recording error as appropriate, and labeling random and systematic errors).
d) summary of your data with appropriate charts, graphics, spreadsheets, etc., drawing on best practices in science and math for data display and analysis
e) statement of what the data means and a summary of the implications of the data you collected
   - A possible sentence could be, “Based on our testing data, we found that <trend in data> means <something about the design's success> because <logic of conclusion>.”
f) discussions about *each* implication from the summary of your data

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g) next steps for your design and process from the new information from the test that was conducted

1) Manuvering Test
    a) High Priority
    b) Improvements from our first test included drilling holes into the piping to ensure the piping fills with water. Adjustments include tilting motors to propel the rover vertically. Maneuvering passes the test
2) Microplastics Collectors Test
    a) High Priority
    b) Hot glue from the first test was unsuccessful and moved onto using zip ties to hold the can in the net. This didn't end up working since the water drag was too strong and it ripped the aluminum. Collector did not pass the test
3) Rover Depth 5 Meters
    a) Medium Priority
    b) Rover had trouble sinking during first trial. 2nd trial we created holes in the piping so the rover could sink fully. This test was successful
4) Rov Buoyancy test
    a) Medium Priorty
    b) Rov had too much buoyancy at first and we had to cut some of the pool floaties off. After realizing the water in the pipes was adding too much weight, we had to add more pool floaties to find a neutral buoyancy for the rover. For our final test, we met this neutral buoyancy and it passed the test.
5) Waterproofed
    a) Low Priority
    b) This was one of our first tests in creating the ROV. We soldered and put shrink wrap around all open wires and were able to close them off to any water. This passed the test.